

How well does it decode?

Participant p01 · 2 blocks per protocol · models always tested on data they were not trained on, unless marked "within session"

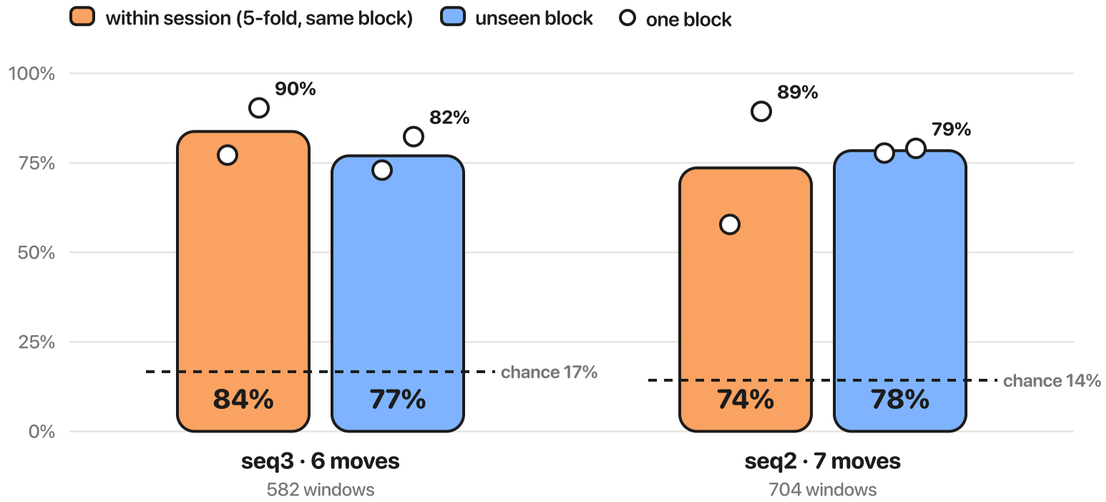
90% **peak window accuracy**
6 moves, within a session

82% **on an unseen block**
77–78% on average · chance 14–17%

1.7 s **median cue → decode**
92% of moves caught (Instant)

a Window accuracy

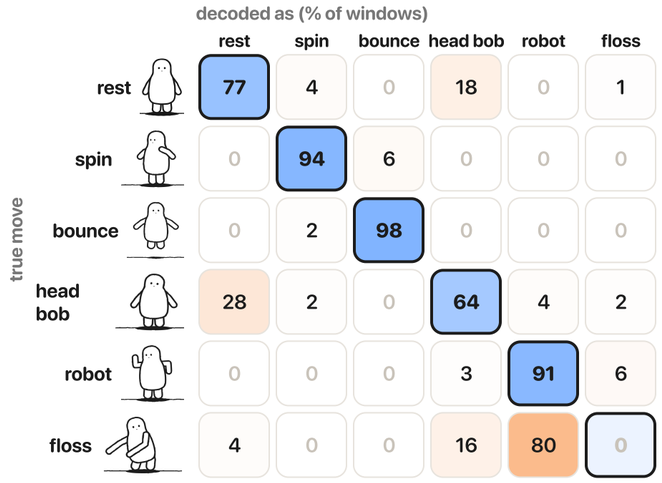
Which move is in a 1.5 s window? Random forest, 92 features.



Up to **90%** within a session and **82%** on an unseen block (77–78% averaged over blocks), 5× chance.

b What gets confused

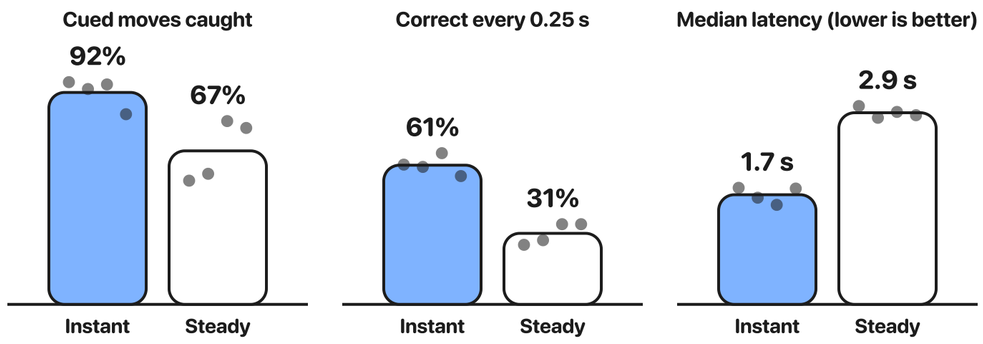
seq3, each block decoded by a model trained on the other block.



Five of six moves are mostly right (64–98%). **Floss is read as robot**: both are arm-swinging moves with similar muscle and motion artifacts.

c Continuous decoding

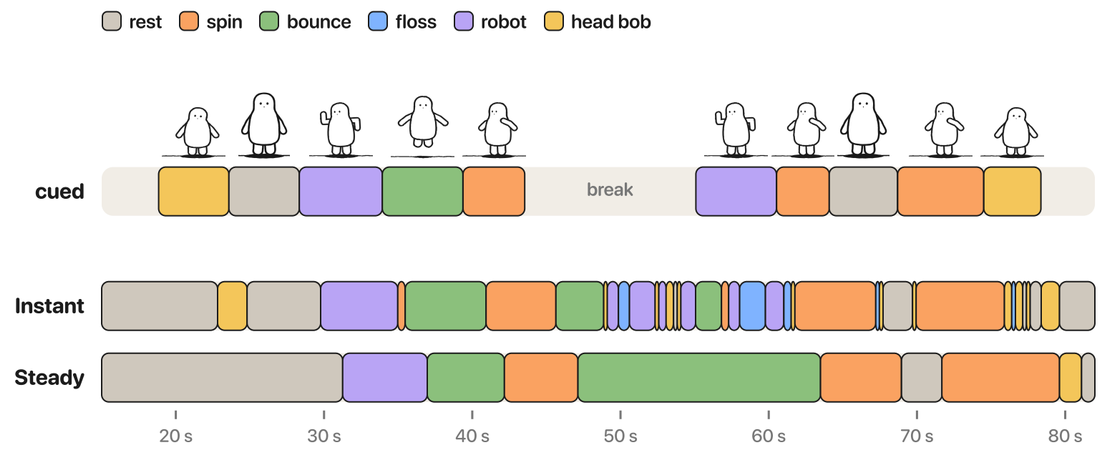
Live-style replay of unseen blocks (seq2 and seq3, 4 replays).



Instant (winner takes all) catches 9 in 10 moves within ~1.7 s of the cue. **Steady** (smoothing, threshold, 0.75 s hold) waits ~1.3 s longer but switches far less often (panel d). Dots: individual replays.

d Cued vs decoded, over time

seq3 block 2, decoded by the block 1 model: two sequences.



In this stretch, 10 cued moves → **39 decoded segments** with Instant, **10** with Steady. Instant follows every change quickly but flickers; Steady gives a cleaner sequence.

Within session: 5-fold cross-validation inside one block, grouped by cued segment (optimistic: same headset fit and session). Unseen block: trained on the other block of the same protocol. Continuous decoding: windows every 0.25 s; a move is caught if it is decoded before the next cue; latency from cue to first correct decode.